

Bubble Coloring to Visualize the Speech Emotion

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Figure 1: Users tend to express positive emotions in a livelier manner, while hiding negative emotions.

ABSTRACT

To explore the emotional effect of the chat bubble's background color on voice messages, we carried out a user survey with Facebook Messenger, WeChat, and KakaoTalk, which use blue, green, and yellow, respectively, as the default color for chat bubbles. We provided the colors in orange, dark red, dark grey, and pale blue when the voice message seemed to be in an excited, angry, sad, or serene mood, respectively. With the exception of the serene, the emotion was intensified through the background color across the three messengers. Concerning willingness to use, the likelihood was reduced, particularly in negative emotions. Based on the empirical evidence, we discussed the potentials and concerns when the method is implemented in voice messaging.

CCS CONCEPTS

• Human-centered computing → Empirical studies in interaction design.

KEYWORDS

voice message, speech emotion recognition, color

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1 INTRODUCTION

Emotions are one of the unique expressions of human nature and, at the same time, one of the elements that researchers endeavor to capture, classify, and visualize in interaction and communication. On one side, technologies have been developed not only to detect emotion through facial expressions, gestures, or movements but also through speech and text[6]. Identifying emotion through verbal and semantic communication has long been an interest of HCI researchers. On the other side, diverse media have been considered to visualize and enhance emotion, such as color, shape, vibration, and images. Among these, color is a compelling option to visualize emotion in communication for robotics[9], graphics[2], interface[3], etc. However, limited studies have demonstrated the synergy from emotion detection to visualization or enhancement.

In this study, we were motivated to apply the emotional characteristics of color to chat bubbles in voice messaging. In particular, we attempted to **visualize emotion from the human voice through color background for messaging**. As both color and human voice convey emotion visually and acoustically, we anticipated the synergy when the two are properly mapped with each other. To be specific, we propose a color alteration service of chat bubbles depending on the voice message's emotional quality. The chat bubbles in different colors may straightforwardly imply a particular emotional state. In this circumstance, we investigated the interaction effect between bubble color and voice emotion. We tried

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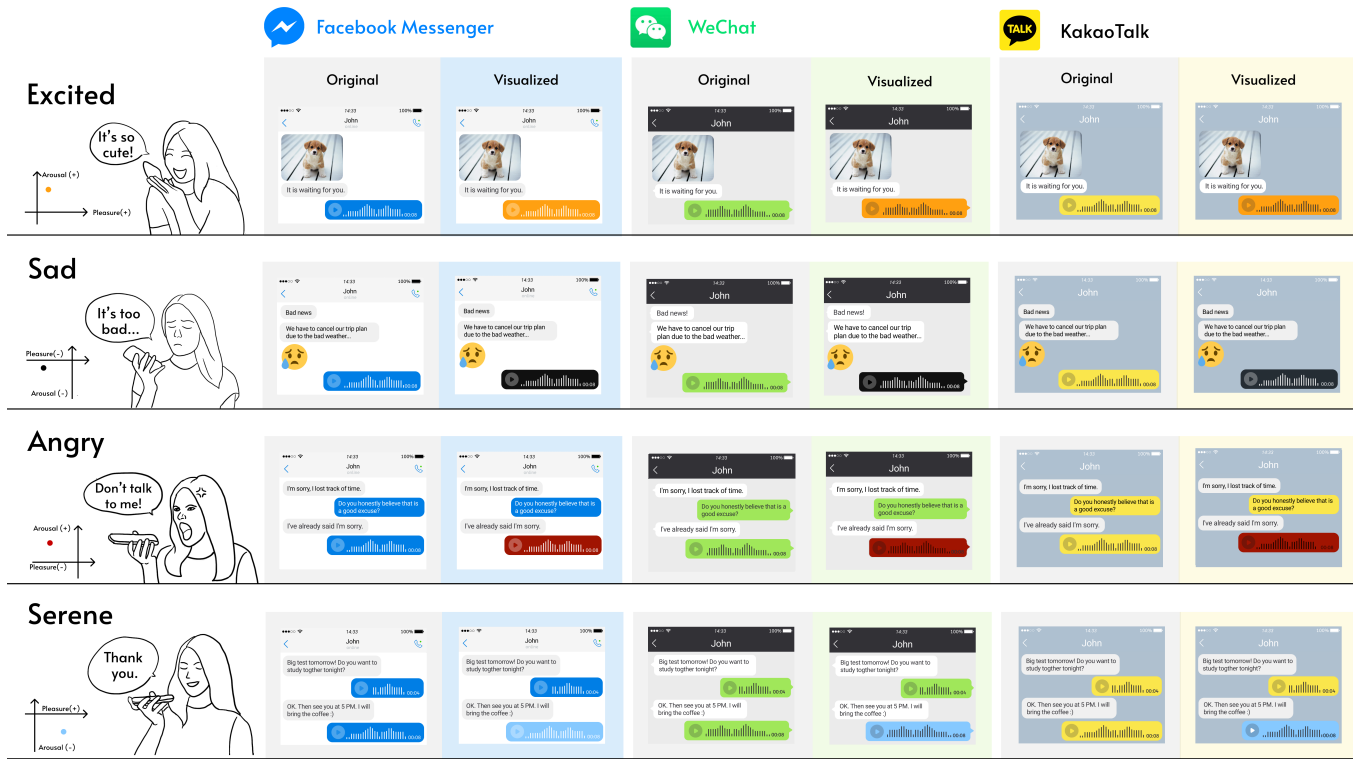


Figure 2: The mapping between the color of the chat bubbles and the four emotions classified from the acoustic characteristics of voice input for three services: Facebook Messenger, WeChat, and KakaoTalk. The mapping referred to previous studies [4, 7, 14, 15] and the default color of the chat bubbles of each service.

to collect empirical evidence from the user study and explore the potential of creating emotional enhancement of voice messages by simply changing the background color.

1.1 Voice instant message and its visualization

Today, instant messaging is a popular choice for daily conversation. According to Statista report[1], WhatsApp and Facebook Messenger ranked top among instant messaging tools globally, while Line, WeChat, and KakaoTalk are popular regionally or locally. Tools have become more advanced to allow multimedia inputs, including audio and video, which are closer to face-to-face conversation than plain text input. Voice messaging has infiltrated the mobile messaging and is available on any major messaging tools.

The recent advancement of speech emotion recognition has made it technically feasible to detect emotion in the human voice[12]. Only recently, however, did researchers start investigating the visualization of emotion in voice messaging. To compensate for the emotional loss of converting voice messages into text, Hu and Xu[8] have suggested the automatic emoji insertion system, and emojis were indeed useful in reducing emotional loss. Additionally, El Ali et al. [5] designed ThermalWear, a thermal display that augments voice perception by thermal stimulation.

1.2 Coloring of chat bubbles

Studies have confirmed the emotional effect of color. From the literature review, we derived tendencies about emotional responses to color. For example, vivid colors are found to be happier and more pleasant, while grayish colors contribute to deactivated or depressive feelings[13]. Based on the previous findings of color and emotional response, we tried to map color and emotions in the context of chat bubbles. When the color of the chat bubble changes, users are supposed to be aware of the change distinctively and understand the target emotion intuitively.

In this study, we considered three representative chat platforms: Facebook Messenger, WeChat, and KakaoTalk. As shown in Figure 2, the chat bubbles are blue, green, and yellow by default. Therefore, the emotion-based color alteration should account for the proper match between color and emotion and the logic of the match, considering the application to the practice.

Regarding emotion, we adopted pleasure and arousal dimensions to conceptualize the emotional quantity of both colors and chat bubbles. Pleasure and arousal are the first two independent axes of emotion and create a two-dimensional emotion space, according to the circumplex model[11], as the two axes cross each other at a right angle. Then, we identified four quadrants and labeled them excited, angry, sad, and serene.

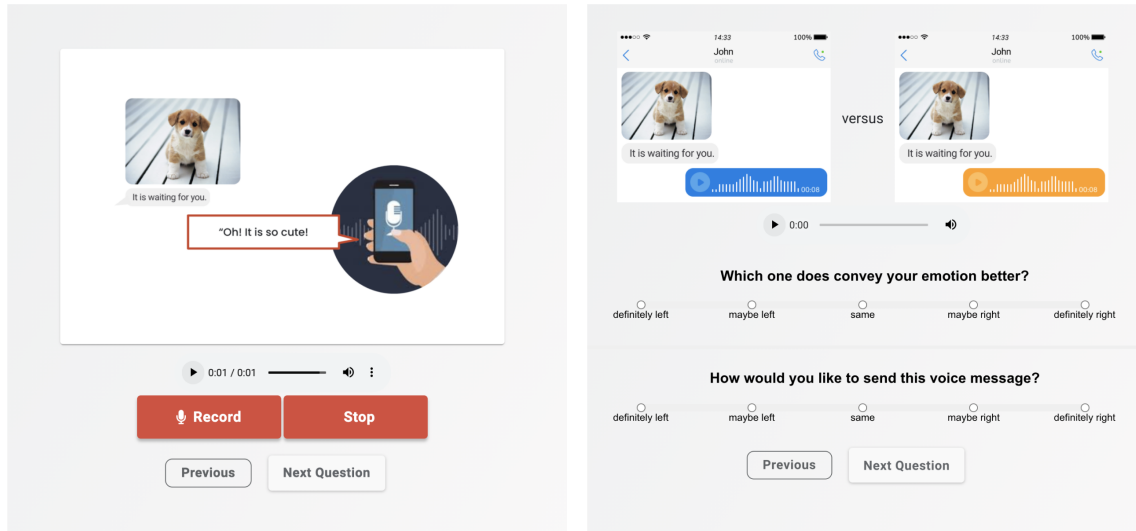


Figure 3: The interface design for the user study. (Left) A conversational scenario targeting excited emotion. Participants were supposed to check the conversation and record the given message emotionally. (Right) A pairwise evaluation survey on the next page. Participants could replay their voice and compare the service with the current service regarding emotional conveyance and willingness to use.

Concerning the emotional characteristics of color, it is hard to conclude one single trend. Nonetheless, studies have replicated common tendencies concerning culture and gender; for instance, red is associated with anger, yellow with excitement, blue with calmness and serenity[4, 7, 10, 14]. This study adopted previous findings while maintaining perceptual distinction from the chat tools' default color.

In other words, we intended to visualize the default chat bubbles in neutral emotion and apply different colors to the chat bubbles of non-neutral emotions, in this case, excited, angry, sad, and serene. As presented in Figure 2, we altered the color of each emotion's bubble background. Orange, dark red, dark grey, and pale blue were selected for excited, angry, sad, and serene, respectively. We also maintained the messenger background color, the text in the bubble, and the voice signal graph. Some revisions were made to the sad emotion for legibility.

2 USER STUDY

2.1 Materials

We prepared a test platform in which participants could experience the service as the message sender and evaluate it compared to the default ones. As mentioned above, we classified the voice into four emotions. One conversational scenario was created for each emotion, allowing participants to clearly understand the conversation atmosphere and the sender's emotion. Furthermore, the platform enabled them to record and replay their voice and feedback on the service.

2.2 Method

We conducted a user study to investigate the emotional effect of the colored chat bubble for voice messages. Participants were required to act as the message sender. They had to check the provided

conversation first and understand the sender's emotion. Then, they were supposed to record a provided voice message that conveyed a particular emotion. The participants were able to retrieve the recorded message and try again until they were satisfied with the recording. After the voice recording stage, the participants were presented with a pair of message screens on the next page. As shown in Figure 3, the default screen was placed on the left, and the chat with the colored bubble was on the right. The participants were required to evaluate the chats concerning two criteria: 1) emotion conveyance and 2) willingness to use. We presented a bipolar scale underneath each criterion in a Likert scale, where -2 referred to prefer default version definitely, 0 referred to the same, and +2 referred to prefer the colored bubble version definitely.

The study was carried out individually and took around 30 minutes to complete. When finished, the voice recordings were archived together with the survey answers and saved on the server. A post-interview was also conducted to help us understand their experience and reasons behind the survey scoring.

2.3 Participants

All participants voluntarily participated in the study and were paid 5 dollars. A total of 25 university students participated, made up of 9 men and 16 women. Their average age was 26.68 years, with a standard deviation of 3.25 years. Although their mother language was diverse, including Korean, Chinese, Russian, and English, they were fluent in English. They were also familiar with the three service platforms and had experience in voice messaging.

3 RESULTS AND ANALYSIS

We aimed to investigate the colored chat bubble's emotional effect on voice messages over different service platforms through the user

Table 1: The mean and standard deviation of user assessments of emotion conveyance and willingness to use for the color bubble of four emotions. (N = 25)

Emotion	Emotion conveyance		Willingness to use		Mean difference	t^b
	Mean(SD)	t^a	Mean(SD)	t^a		
<i>excited</i>	0.83(0.96)	7.43*	0.80(1.95)	6.58*	-0.03	0.36
<i>angry</i>	1.43(0.87)	14.20*	0.63(1.26)	4.31*	-0.80	5.79*
<i>sad</i>	1.33(0.83)	14.00*	1.00(1.03)	8.55*	-0.33	3.72*
<i>serene</i>	0.17(1.17)	1.29*	0.15(1.28)	0.99*	-0.02	0.38
^a t scores from one sample t -test to statistically examine the mean difference with 0.						
^b t scores from paired sample t -test to statistically compare the mean difference between two criteria.						
* $p < 0.05$						

study. We collected the participants' assessment of the colored bubble service and the post-interview transcripts about their thoughts behind it. Therefore, we conducted a quantitative analysis for the survey and a qualitative analysis for the interview.

3.1 “Does the bubble color enhance the emotion in the voice message?”

On average, the participants found the bubble coloring more emotionally intensive, supporting our assumption. The emotional conveyance of the four emotions were evaluated as 0.83 (SD: 0.96), 1.43 (SD: 0.87), 1.33 (SD: 0.83), and 0.17 (SD: 1.17) for excited, angry, sad, and serene, respectively. The dark red background for angry received the highest score, followed by dark grey for sad and orange for excited. As the null hypothesis is no difference between default and colored bubble, we performed a one-sample t -test to examine whether the scores were statistically different from 0.

From the analysis, colored bubble chats statistically received a higher score than 0 for all four emotions, excited [$t(74) = 7.43$, $p < 0.05$], angry [$t(74) = 14.2$, $p < 0.05$], sad [$t(74) = 14$, $p < 0.05$], and serene [$t(74) = 1.29$, $p < 0.05$]. The mean scores together with the standard deviation were shown in Table 1.

From the statistical analysis, all mean differences were significant at an alpha level of 0.05, indicating that the emotional conveyance was overall intensified. However, we observed that the effect on serene was marginal, despite the statistical significance. In other words, coloring the background of the serene voice message in pale blue did not enhance the emotional quality as much as the other colors did.

3.2 “Would you like to color the bubble in your conversation?”

Although the colored bubble enhanced the user's emotion, willingness to use it was a different issue. Therefore, we performed a one-sample t -test to examine whether the corresponding ratings were statistically different from the current service for the willing to use criteria. As shown in Table 1, the results showed that colored bubbles scored statistically higher for all the emotions, excited [$t(74) = 6.58$, $p < 0.05$], angry [$t(74) = 4.31$, $p < 0.05$], sad [$t(74) = 8.55$, $p < 0.05$], and serene [$t(74) = 0.99$, $p < 0.05$]. It indicated

that participants were likely to use the colored bubble, rather than the default version, during chatting to visualize their emotions. To be specific, sad and excited received a relatively higher average score.

We performed a paired sample t -test to examine the participants' evaluations of the two criteria. The results yielded no significant differences between the two criteria in the case of excited and serene. However, regarding angry and sad, the mean score of willingness to use significantly decreased: angry [$t(74) = 5.79$, $p < 0.05$], sad [$t(74) = 3.72$, $p < 0.05$]. It suggested that the negative emotions may be less likely to be explicitly visualized than the positive emotions, even though they can intensify the emotional expression.

3.3 “Do these differ by service platform?”

To compare the effect of the emotion and the service on the mean rating score of two criteria, we employed a two-way ANOVA. We used pairwise comparisons as post hoc test and Bonferroni correction was applied to the p -values for multiple comparisons.

For emotion conveyance, there was a statistically significant main effect of emotion ($F[3,288] = 27.43$, $p < 0.05$) and a two-way interaction between the emotion and service ($F[6,288] = 2.51$, $p < 0.05$). However, we did not find any significant main effect of service ($F[2,288] = 1.37$, $p > 0.05$). This indicated that the service was only effective when combined with emotion, confirming our assumption of the interaction effect of color combinations of the default color and bubble color. As shown in Figure 4, the post hoc test revealed that for serene, almost all pairwise comparisons were statistically significant ($p < 0.05$). That is, the pale blue could not represent serene feelings across all platforms, while the other three bubble colors' emotional conveyance reached a considerable level of agreement on all platforms.

For willingness to use, although we did not find statistically significant main effect of service ($F[2, 288] = 0.04$, $p > 0.05$), there was a statistically significant main effect of emotion ($F[3, 288] = 7.61$, $p < 0.05$) and a two-way interaction effect of emotion and service ($F[6, 288] = 2.62$, $p < 0.05$). Generally speaking, willingness to use also followed the tendency of emotion conveyance, affected by the interaction effect between service and emotion, but not by service.

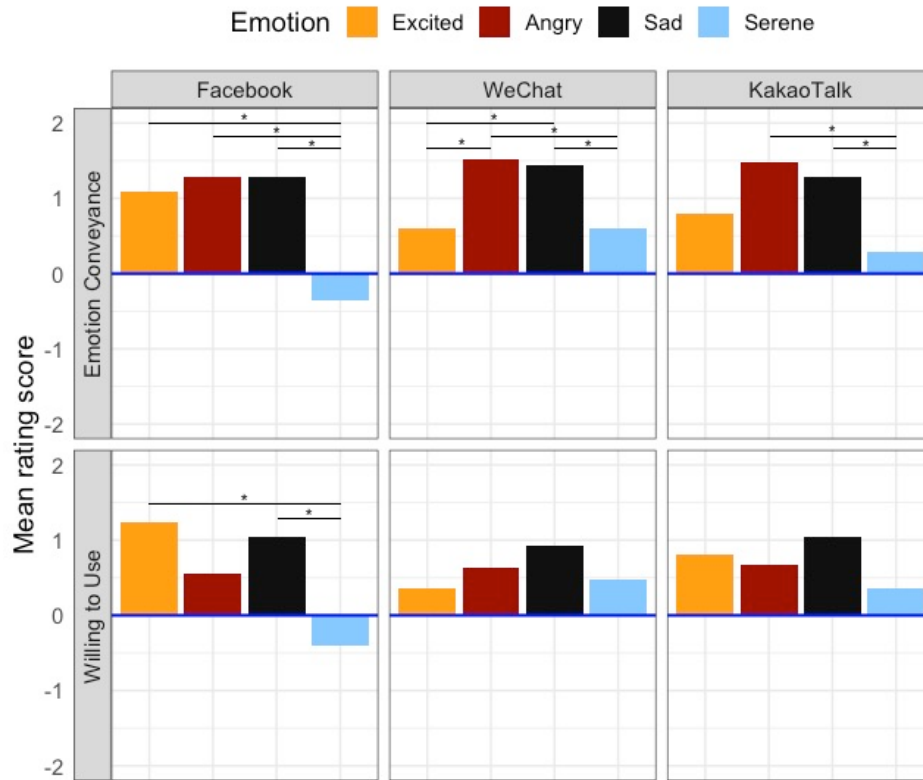


Figure 4: Average scales of the user assessments of emotion conveyance and willingness to use for the four emotions on three different service platforms: Facebook Messenger, WeChat, and KakaoTalk.

3.4 “What else did you find out during the survey?”

After the survey, we conducted individual interviews to collect thoughts and feedback beyond the two questions included in the survey. We gathered meaningful comments to develop the bubble coloring into a proper algorithm. Three findings are summarized below.

3.4.1 My choice to apply color bubble. The system should allow the user to apply the colored bubble or not. Our participants stated that voice messages were typically for informal daily communication; they mostly receive or send voice messages to family and close friends. There were also some situations in which they had no other choice but to send a voice message, such as while running or driving. However, based on the relationship with the receiver, their attitudes to visualize the emotion changed. For example, sending a colored voice message to unfamiliar people was considered inappropriate by most participants. Therefore, using the colored bubble should be an option for the users rather than an automatic algorithm attached to the messenger.

3.4.2 Confused color alterations. In this study, orange, dark red, dark grey, and pale blue were selected for excited, angry, sad, and serene, respectively. Although these color selections referred to previous studies, many participants were confused by some of the

colors to reflect the targeted emotions. Pale blue was the most arguable one. Some participants reported that blue was associated with feeling blue semantically, which reminded them of feeling sad. Some even suggested using light yellow or light orange for these serene, calm, and relaxed emotions. Moreover, some participants thought orange was also close to the meaning of danger. Although the colored bubble intuitively expresses any specific emotion, the accuracy of the emotion conveyance was still an issue. Furthermore, the conversation itself was at least two persons. These differences in perceived emotion could lead to confusion in communication, which should be better considered.

3.4.3 Harmony with default bubble color. Originated from the brands’ logos, each platform uses completely different default colors that certain emotional states. Furthermore, many participants talked about their impression of the colored bubble regarding the platform’s default color. Some participants reported that the orange of excited looked much more excited in Facebook Messenger than in WeChat, caused by the hue contrast between blue and orange. Furthermore, when a pale blue appeared in Facebook Messenger, where the chat bubbles were blue by default, the pale blue reminded them of failure in message delivery or loading. These all indicated that when using the colored bubble, participants paid attention to color harmony between the default color and the altered colors.

Moreover, not all four color alterations were presented to participants in the service interface for the first time. This made them fail to identify the difference in color in some cases. Therefore, when the colored bubble is developed into service, the color alterations should be well communicated to the users.

4 CONCLUSION

This paper proposed a service prototype that employed a colored bubble to convey the emotional states detected from voice messages in instant messenger tools. To examine the effectiveness of this approach, we conducted a user study with 25 university students. The results demonstrated the feasibility of using bubble color to express different emotions. We observed that the bubble color visually indicated emotion and intensified a voice message's emotional state before listening. Although bubble color can enhance the voice message's emotional expression, whether to use it or not was another issue. In the case of excited and sad, participants preferred to use a color bubble to visualize their emotions for voice messages, whereas for angry, although the dark red effectively enhanced the emotional state, participants were less likely to want to visualize their emotion. We believe that bubble coloring has a realistic potential in messenger service and can facilitate more studies in the research area of emotion detection and visualization.

However, there were some limitations in this preliminary study. First of all, four emotions were addressed, concerning the four quadrants in the arousal and pleasure axes. More specific emotions can be considered in future work with better color matching to visualize emotion in daily communication. Second, the cultural differences of participants could influence results. Their first language may also influence their emotional perception and expression concerning color[10]. Furthermore, we could not observe user behaviors in natural situations. Sending a voice message was still private for many people. Because all the conversations were pre-defined, and the participants knew that conversations would be analyzed, it was difficult for them to experience the service naturally.

Nonetheless, our findings indicated that the colored bubble could contribute to users' emotional expression in message communication and created a lively atmosphere during chatting. In the future,

we plan to implement a colored voice messaging system with a speech emotion recognition model in real messenger applications and expected to find more values through a long-term study.

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